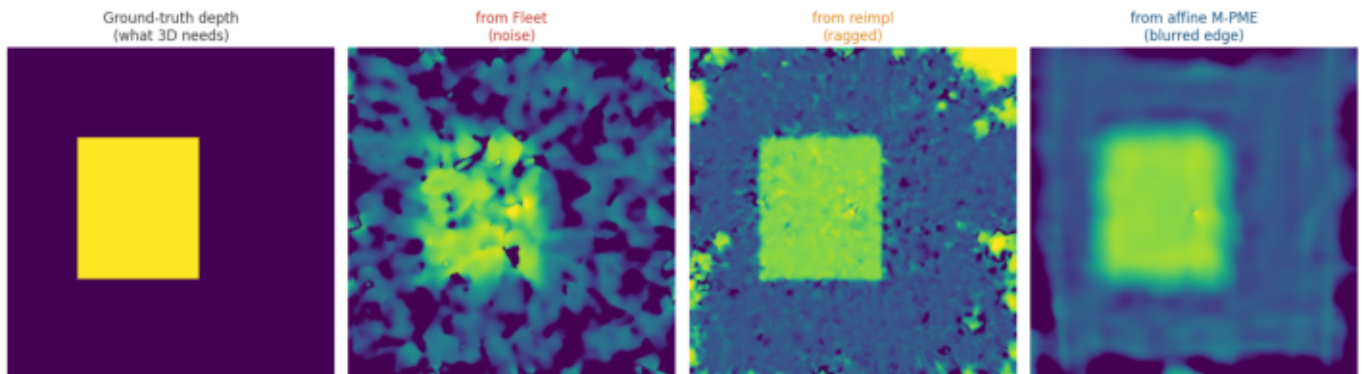


The Affine Ceiling

Formal motion estimation is sharp on affine motion — and blind at depth.



Depth reconstructed from each method's optical flow under a perspective Z-dolly. Ground truth is crisp; every phase-based method — including the affine state of the art — blurs, frays, or destroys the depth boundary. That boundary is what 3D reconstruction is made of.

Motion estimation is the front end of 3D reconstruction. This note measures how well formal, non-probabilistic phase-based methods actually recover it — against pixel-exact ground truth, across three implementations. The result splits cleanly in two: they solve affine motion to sub-pixel accuracy, and they cannot recover usable depth.

Two ways to estimate motion

Learned monocular models — the neural depth and matching networks now driving the field — are remarkably capable. But they are probabilistic by construction: no error bound, no repeatable output, no provable worst case. The same scene in different light returns different numbers. For a photo app, fine. For a system under functional-safety standards — ISO 26262, DO-178C, IEC 62304 — it is disqualifying.

This work takes the opposite stance: formal, non-probabilistic estimation, derived from signal theory rather than trained from data. Every output is reproducible and analyzable. The open question is not whether it can be trusted — it is whether it is accurate enough. Specifically: accurate enough for 3D.

02 – THE SINGLE-SCALE LIMIT

A limit you can derive on paper

Phase-based estimation reads displacement from the phase shift of a band-pass filter. The Fourier shift theorem makes that shift linear in the displacement:

$$\Delta\phi = 2\pi k \delta \quad \rightarrow \quad \delta_{\max} = \lambda/2$$

phase is unique only in $(-\pi, \pi]$;
past half a wavelength it wraps

Beyond half a filter wavelength the phase aliases and the answer is silently wrong. Fleet & Jepson characterized this in the early 1990s [1][2]. A single-scale method therefore has a hard ceiling — and it is small. The question is whether that ceiling is fundamental. It is not: a multi-scale pyramid pushes it away, as we show next.

From a hard ceiling to none

- Fleet (single-scale). One octave band, measured local frequency, singularity masking. Bounded to $\delta < \lambda/2$ — our reference for the limit.
- Reimpl (no paper). Multi-scale Gaussian pyramid, per-pixel translation, rewarping. Built before reading the source paper.
- M-PME (faithful). The 2026 method of Li et al. [5]: a Gaussian pyramid with an AFFINE 6-parameter motion model fit over a window (Farnebaeck-style), confidence weighting, four Gabor directions, coarse-to-fine. Scale and rotation ARE affine — it models them exactly.

04 — AFFINE MOTION IS SOLVED

Sub-pixel on large scaling

We warp a broadband texture by an exactly known scaling (a fronto-parallel plane under a Z-dolly is precisely a scaling) and sweep the magnitude. Single-scale fails past ~ 11 px; both multi-scale methods stay sub-pixel; the affine M-PME is cleanest.

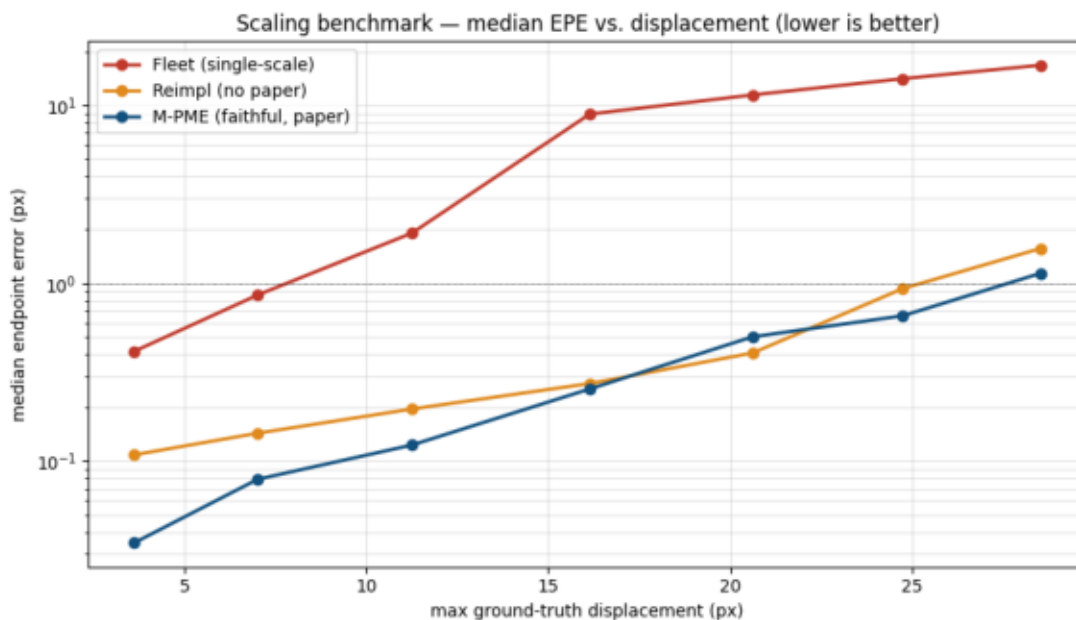


Fig. 1 Median EPE vs. displacement (log scale). Affine motion is solved to sub-pixel accuracy across the range.

So the intuitive story — 'formal methods only do small motion' — is false. A multi-scale, affine, coarse-to-fine method solves large scaling and rotation. The ceiling is not motion size. It is the affine assumption itself — and real scenes break it.

Real depth is not affine — and it shows

A real scene is not a single plane. Move the camera along Z over a near object in front of a far background, and the optical flow is depth-dependent, with a DISCONTINUITY at the object's edge. That violates the affine-per-window assumption every method here relies on. Ground truth (flow AND depth) is exact, built from the depth map and the known camera step.

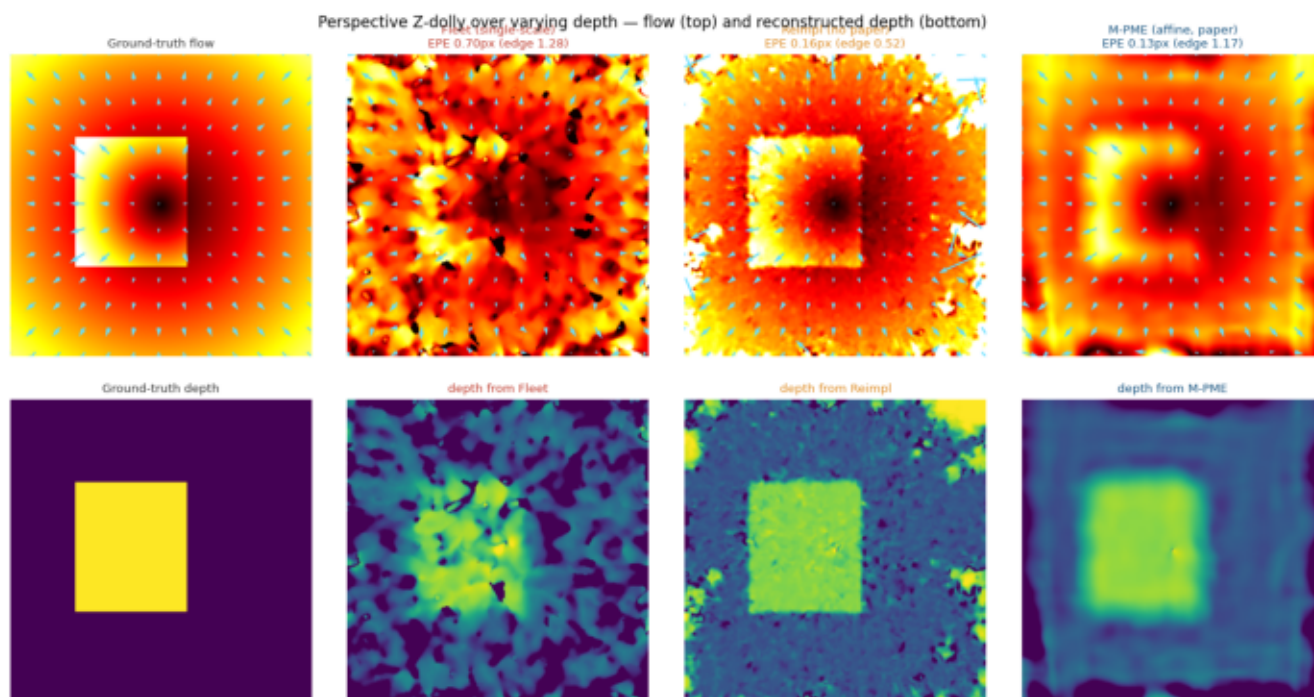


Fig. 2 Perspective Z-dolly over a two-plane scene. Top: recovered flow. Bottom: depth reconstructed from that flow. GT depth is a crisp step; Fleet returns noise, the reimplementation frays, the affine M-PME blurs the boundary away.

- On the smooth (locally-affine) plane regions all methods do fine — median EPE 0.13-0.70 px.
- At the depth edge they break: EPE jumps 4-9x. The affine M-PME is WORSE at the edge (1.17 px) than the simpler reimplementation (0.52 px) — its large window averages across the discontinuity.
- The reconstructed depth is unusable: noise, ragged edges, or a blurred step where 3D needs a crisp one.

Sharp on affine. Blind at depth.

Every method here is sharp on affine motion and blind at the depth boundary. For 3D reconstruction, that boundary is the whole point.

The two halves of this benchmark bracket the state of formal motion estimation. On affine motion — translation, rotation, scale — it is genuinely solved, to sub-pixel accuracy, at large magnitude. On the non-affine, per-pixel, discontinuous motion of a real 3D scene, it produces flow that looks plausible in aggregate but destroys exactly the depth edges that reconstruction depends on.

That is the gap. The probabilistic models that handle real scenes cannot be certified; the formal methods that can be certified cannot yet resolve depth. Neither is usable for safety-critical 3D as it stands.

Outlook

We are building a formal, non-probabilistic method designed to stay sharp exactly where these blur — crisp at the depth boundary, accurate enough per pixel to drive 3D reconstruction. This note is the baseline it will be measured against.

METHOD NOTE

Controlled synthetic scenes with pixel-exact ground truth (broadband texture; scaling and two-plane Z-dolly). EPE is the median over a central crop, excluding the focus-of-expansion core. M-PME is our faithful reimplementation of the described method. All figures reproducible from source.

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